Ahmed Ikram | Data Engineer

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Final-year BSc Computer Science (Data Science & AI), Dundee (predicted First, June 2027). Built batch-lakehouse and streaming platforms in Agile/Scrum teams - defining my own report metrics from 156K-row data, gating pipelines others trust. Explain complex ideas simply: mentored 30+ primary pupils one-to-one. Operationally trusted early: solo shop-closing, cash reconciliation, training new staff. Seeking a graduate Data Engineering role from June 2027.

EDUCATION

University of Dundee - *BSc (Hons) Computer Science (Data Science and AI)* *Sep 2023 - Present*

- On track for a First-Class Honours degree. Expected Graduation: June 2027
- Honours Project: Tiny Language Model Laboratory - Reproducible ML Experimentation Platform

  Developing a reproducible experimental platform for generating synthetic sequence datasets, training configurable Transformer language models and tracking metrics, checkpoints and experiment configurations. Designing data and training pipelines that support controlled dataset-size, task-difficulty and model-configuration experiments, with visualisations for training behaviour, memorisation and generalisation.

High School of Dundee *Sep 2017 - June 2023*

- Highers: A Mathematics, A Computing Science, A English, B Chemistry.
- Advanced Highers: Computing Science C.
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

TECHNICAL SKILLS

Languages & Stores: Python · SQL · T-SQL · Java · C · Bash · PostgreSQL · Redis · Alembic · ChromaDB

Orchestration, Scheduling & Transformation: Apache Airflow · Apache Kafka · dbt · Apache Spark / PySpark · systemd timers / cron · Pydantic · MLflow

Streaming & Warehousing: OpenLineage · Marquez · ClickHouse · Delta Lake · BigQuery · Databricks (DLT, Unity Catalog) · Azure SQL · Power BI · Excel

Data Quality & Observability: Great Expectations · Grafana · Prometheus · Azure Monitor · Cloud Monitoring · CloudWatch · StatsD · Evidently AI · OpenTelemetry (Jaeger) · Query Optimisation · Schema Evolution · Partitioning

Cloud & DevOps: AWS (EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager, DynamoDB) · Azure (Databricks, Azure SQL (SQL Server), ADLS) · Google Cloud (GCS, BigQuery) · Terraform · Docker · GitHub Actions · Linux

PROJECTS

W3C Web Logs ETL Pipeline - ADLS to Power BI

Airflow · Databricks · dbt · PySpark · Azure SQL · Delta Lake · Terraform · Power BI · Marquez · Grafana · Prometheus · CI/CD

- Architected a Databricks DLT medallion in PySpark (Bronze & Silver in DLT; gold star schema via dbt) on serverless compute: 153,380 rows with 3 Silver rows dropped through 10 DLT expect-or-drop gates; Silver consumes Bronze via a checkpointed Delta Change Data Feed stream (readChangeFeed).
- Consolidated 7 scalar GeoIP PySpark UDFs into a single struct-returning UDF (reads MaxMind databases from Unity Catalog Volumes); runs ~3.5x faster and enriches 153,377 Silver rows to 99.99% known-country coverage.
- Owned a skew-resilient PySpark range-backfill job re-applying the full Silver enrichment to any Bronze date range: broadcast crawler-IP join, salted repartition over (log_date, salt), idempotent Delta replaceWhere skew 14x → 1.5x.
- Modelled the Gold layer as a Kimball star schema: 16 dbt models (SCD1; SCD Type-2 on GeoLocation via T-SQL MERGE) on hashed natural keys, 18 macros dual-compile PostgreSQL ↔ T-SQL; 121 dbt tests catch 213 duplicate rows on a 12-component deduplication key.
- Orchestrated 2 Airflow DAGs with dataset-triggered decoupling of ingest from transform; OpenLineage → Marquez records cross-engine lineage (DLT → Azure SQL → dbt) beyond Unity Catalog's per-engine view, including dbt test outcome facets.
- Backed a 7-page Power BI report (156K requests, 88 countries) with a freshness probe and 3 Grafana dashboards (23 panels, 8 Prometheus rules, 2 Azure Monitor alerts).
- Provisioned with a 2-part Terraform estate (30+ Azure resources, OIDC federation, rollback workflow), validated by 750 tests across 5 tiers including dual-dialect dbt CI and an OIDC authenticated Azure CD workflow.

WikiStream Real-Time Streaming Analytics

Python · ClickHouse · Grafana · BigQuery · Great Expectations · Terraform · Google Cloud · Pydantic · CI/CD

- Live Streamed Wikipedia's EventStreams feed into self-hosted ClickHouse - 58.9M+ events in 3 days with 2,719 events/s real peak; Pydantic schema-on-write rejects malformed events to a dead-letter table.
- Guaranteed at-least-once delivery across reconnects and kills via durable cursor & deduplication ring buffer; provd with a live SIGKILL-mid-insert test (zero loss, zero duplicates).
- Cut dashboard latency 15x at p50 (13x at p99), ~200x fewer rows scanned, with three materialised views; held 5,655 events/sec (2x real peak) in a 10x synthetic burst with zero drops.
- Automated hourly ClickHouse → BigQuery warehouse loads (idempotent staging & MERGE, exact-sum parity check every run), plus hourly GX data-quality checks; 2-year KPI retention / 90-day raw-sample retention.
- Built a daily gold table - day-over-day growth - via a scheduled BigQuery rollup, with partition filters, clustering, and a cost dashboard to bound scan spend.
- Terraform/WIF GCP stack with an itemized ~$41/month run-rate; CI gate: 161 tests (live ClickHouse), 100% consumer-core coverage, schema-drift & parity contracts.

ATM Log Aggregation, Anomaly Detection & Diagnostics - Event Pipeline for NCR Atleos

Apache Kafka · FastAPI · PostgreSQL · Redis · ChromaDB · MLflow · Docker · Terraform · AWS · SageMaker · CI/CD

- Owned a distributed multi-host event pipeline: Kafka (KRaft) at-least-once delivery with idempotent deduplication (Redis SET+TTL, LRU fallback), Redis Stream DLQ with exponential backoff, ~100 msg/s over 2.5M events across 7 log sources, with OpenTelemetry-tracing via Kafka propagated headers, surfacing exact per-decision USD cost.
- Built a true Hybrid RAG Agent with LangGraph: a streaming consumer feeds logs into ChromaDB through semantic chunking and local embedding, retrieval is dense + BM25 → RRF k=60 via search_knowledge and MCP tooling.
- Stored ATM logs in a PostgreSQL JSONB lake - a new source is a parser, not a migration; 15 indexes (6 composite) held sub-100ms queries past 4M+ rows.
- Tracked both training runs and live production anomaly detection runs in MLflow with champion-alias promotion, so ML consumers read a gated artifact, cross-checked against a live SageMaker endpoint.
- Provisioned the AWS estate in Terraform (multi-AZ VPC, ECS Fargate, RDS, S3, CloudFront, 4 least-privilege IAM roles, state locked in DynamoDB versioned in S3) validated by 1,846 tests.

StockLens Feature & Retraining Pipeline

Rust · Airflow · AWS · PyTorch · MLflow · Evidently AI · PostgreSQL · Redis · Terraform · CI/CD

- Orchestrated a weekly Airflow retraining DAG: one-time 28M-row Kaggle bulk seed + weekly yfinance refresh → 17-feature build (13 via a Rust/PyO3 engine) → MLflow-tracked 5-seed LSTM training → paired-McNemar gated champion/challenger promotion (>2pp + $p<0.05$).
- Integrated Evidently AI drift monitoring PSI/KS/JSD on live vs training distributions, alerting to CloudWatch before drift reaches the champion - the same quality-gate pattern as the data pipelines, applied to a model artifact.
- Automated champion delivery: the gated 5-model ensemble bundle pushed to S3 at promotion & bootstrapped to Fargate for zero-copy inference, with PostgreSQL model registry and prediction logs retained 90 days for offline replay.

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations	F*eb 2025*
Foundations of Azure AI: Concepts, Capabilities, and Implementation	*Oct 2024*
AWS Academy Graduate - Cloud Foundations	*Mar 2024*

EXPERIENCE

Beauly Mini Market, Dundee, Scotland

Retail Assistant — *Sep 2023 - Present*

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker — *June 2021 - Sep 2023*